Who Made This Image? Beyond Detection: Identifying the Generators of AI-Generated Images

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Abstract

Recent advances in AI image generation have increased both their practical utility and their misuse in applications such as phishing, scam advertisements, and forged documents. In these contexts, just distinguishing real images from fake ones is insufficient; identifying the specific AI generator responsible for an image enables proactive defense, forensic analysis, and accountability. In this work, we address the task of AI image generator identification by analyzing generator-specific visual characteristics in synthetic images. We conduct experiments on the LSUN Bedroom dataset, which contains images produced by ten AI generators spanning both GAN-based and diffusion-based models, with explicit generator annotations. To capture complementary visual cues, we systematically explore different combinations of pretrained vision representations operating at multiple visual domain levels. Specifically, we examine the combination of CLIP for high-level semantic and stylistic information, DINOv3 for fine-grained visual structures, and a frequency-domain vision encoder trained with Masked Frequency Modeling vision encoder to capture low-level artifacts. We further adopt Visual Prompt Tuning as a parameter-efficient fine-tuning strategy under limited data and computational resources. Our experiments aim to identify effective and efficient combinations of these representations for AI image generator detection. The code is available at `https://github.com/lucinnnal/ai_forensics`

1 Introduction

Recent years have witnessed remarkable progress in AI-based image generation models, including GANs and diffusion models, to the extent that their outputs are often indistinguishable from real images by human observers. While these advances have enabled numerous creative and practical applications, they have simultaneously raised serious concerns regarding misuse, such as phishing attacks, scam advertisements, forged documents, and large-scale misinformation campaigns. As a result, AI image detection, which aims to distinguish AI-generated images from real ones, has become an active research area.

A large body of prior work has reported impressive performance on binary real/fake classification, often achieving near-saturated accuracy on established benchmarks. However, such results do not necessarily imply that the models have learned the intrinsic characteristics of AI-generated images. Instead, high performance may stem from exploiting dataset-specific biases, compression artifacts, resolution discrepancies, or generator-dependent shortcuts that do not generalize beyond the evaluation setting. Consequently, the question of what these models truly learn remains largely unanswered.

Motivated by this limitation, we argue that binary AI image detection is an overly permissive task for evaluating a model's representational capacity. If a model genuinely captures the distinctive properties of AI-generated images, it should be capable of more than simply answering "Is this image real or fake?". In particular, it should be able to identify which generator produced the image. This observation motivates the task of AI image generator identification, a substantially more challenging and discriminative problem. Different image generators differ in architectural design, upsampling strategies, noise injection and removal procedures, denoising schedules, samplers, training data distributions, and normalization schemes. These differences manifest as generator-specific visual signatures, including spatial patterns (e.g., texture regularities, boundary inconsistencies, local structural artifacts) and

frequency-domain characteristics (e.g., abnormal energy distributions, repetitive high-frequency components, or upsampling-induced patterns). As a result, generator identification requires models to capture fine-grained, generator-dependent cues, rather than relying on coarse real/fake signals. Therefore, it serves as a more rigorous testbed for assessing whether AI image detection models truly learn meaningful and generalizable representations.

In this work, we study the task of multi-class AI image generator identification, where the goal is to identify the specific generator responsible for a given AI-generated image. We conduct our experiments on the LSUN Bedroom dataset containing images generated by ten distinct models, covering both GAN-based and diffusion-based approaches, with explicit generator annotations. This setting enables a controlled and systematic analysis of generator-specific visual characteristics. To investigate what types of representations are most effective for this task, we explore a range of pretrained vision models that capture complementary visual cues at different levels. First, we consider CLIP, which provides powerful high-level semantic and stylistic representations learned from large-scale image–text pretraining. Such representations are well suited for capturing global appearance and style biases that may differ across generators. Second, we examine DINOv3, a self-supervised vision model known for encoding fine-grained local structures and patch-level consistency, making it sensitive to subtle spatial artifacts in generated images. Third, we incorporate a frequency-domain vision encoder trained via Masked Frequency Modeling (MFM), which explicitly targets low-level artifacts by reconstructing masked frequency components, thereby emphasizing generator-specific signals in the frequency domain.

Beyond representation choice, we also consider practical constraints on data and computation. Rather than fully fine-tuning large pretrained models, we adopt Visual Prompt Tuning, a parameter-efficient adaptation strategy that introduces a small set of learnable prompt tokens while keeping the backbone frozen. This design enables efficient domain adaptation while significantly reducing training cost, making generator more feasible in realistic settings. By systematically combining semantic-level representations (CLIP), fine-grained visual structures (DINOv3), frequency-domain artifacts (MFM), and prompt-based adaptation, we aim to answer a fundamental question: What types of visual representations truly capture the distinctive characteristics of AI image generators?. We summarize the key contributions as follows:

- We move beyond conventional real/fake detection and formulate AI image generator identification as a more stringent evaluation task for AI image detection models.
- We conduct a systematic study of pretrained representations operating at multiple visual domain levels-semantic, structural, and frequency-based—and analyze their effectiveness for generator identification.
- We demonstrate that combining complementary representations with parameter-efficient visual prompt tuning provides a practical and effective approach for identifying the generators of AI-generated images.

2 Related Work

2.1 AI-Generated Image Detection

CLIP-based semantic representations. It has been shown that pretrained vision–language models, particularly CLIP, provide highly transferable embeddings for detecting AI-generated images [1]. By operating directly in CLIP feature space with lightweight classifiers and minimal training data, their method generalizes well to unseen diffusion models and commercial generators. This line of work demonstrates that high-level semantic representations capture generator-agnostic cues beyond traditional low-level forensic traces. However, CLIP features are inherently general-purpose and may encode substantial task-irrelevant semantic information. To address this limitation, there has been an approach which integrates a Variational Information Bottleneck on top of CLIP embeddings to suppress irrelevant semantic factors and retain task-relevant signals for real–fake discrimination [2].

Frequency-domain and spectral learning methods. An alternative line of research focuses on frequency-domain cues introduced by generative models. A self-supervised spectral learning approach has been introduced to characterize the spectral distribution of real images, framing generated images as out-of-distribution samples [3]. This approach achieves strong generalization across unseen generators and resolutions, suggesting that invariant spectral properties of real images provide a robust detection signal.

Hybrid semantic–spectral approaches. Motivated by the complementary strengths of high-level semantic representations and low-level statistical cues, a hybrid detection framework has been proposed that integrates CLIP-based embeddings with frequency based features [4]. Experimental results indicate that jointly leveraging semantic and spectral information leads to improved robustness in detecting AI-generated images.

2.2 Visual Representation Learning for AI-Generated Image Analysis

CLIP CLIP (Contrastive Language–Image Pre-training) [5] adopts a dual-encoder architecture that separately encodes images and texts, aligning their representations in a shared embedding space through large-scale contrastive learning on image–text pairs. The training objective increases similarity between matched image–text pairs while suppressing similarity for mismatched pairs within a batch. Through natural language supervision, CLIP learns transferable image representations that are strongly coupled with high-level semantic concepts, without relying on fixed dataset-specific labels.

In the context of AI-generated image analysis and generator identification, CLIP provides high-level visual representations that capture semantic coherence, global composition, and stylistic biases such as texture, illumination, color distribution, and layout. These properties allow CLIP to encode generator-dependent tendencies in visual plausibility and semantic consistency, making it a widely adopted backbone for detecting and analyzing synthetic imagery.

DINOv3 DINOv3 [6] is a large-scale self-supervised vision model designed to jointly achieve global discriminability and high-quality local representations within a single visual encoder. Its training objective combines multiple self-supervised losses defined at both image-level and patch-level, encouraging balanced learning of global structure and fine-grained visual patterns.

DINOv3 relies purely on visual information without any text-based semantic alignment. As a result, it emphasizes visual patterns, structural regularities, and local details rather than semantic cues. Such properties make DINOv3 particularly suitable for capturing subtle local artifacts and structural inconsistencies that may arise in AI-generated images, offering a complementary perspective to semantic-driven representations.

Masked Frequency Modeling Masked Frequency Modeling (MFM) [7] is a self-supervised pre-training approach that operates in the frequency domain, in contrast to conventional masked image modeling methods defined in the spatial domain. MFM transforms images into frequency spectra and performs a mask-and-predict task by masking subsets of frequency components and reconstructing the missing information directly in the spectral space. This frequency-domain formulation mitigates spatial redundancy and enables the model to exploit both high-frequency structural cues and low-frequency statistical patterns.

2.3 Parameter-Efficient Domain Adaptation

Visual Prompt Tuning Visual Prompt Tuning (VPT) [8] is a parameter-efficient fine-tuning strategy for adapting large pretrained Vision Transformer (ViT) models to downstream tasks. Instead of updating all backbone parameters, VPT introduces a small set of learnable prompt tokens that are prepended to the input token sequence. During training, only the prompt tokens (and a lightweight task-specific head) are optimized, while the pretrained backbone remains frozen.

By modifying the input representation rather than the model weights, VPT enables effective domain adaptation with a minimal number of additional parameters. This property has made prompt-based tuning particularly attractive in scenarios with limited labeled data or when adapting large-scale pretrained models to specialized domains, including AI-generated image detection and generator identification.

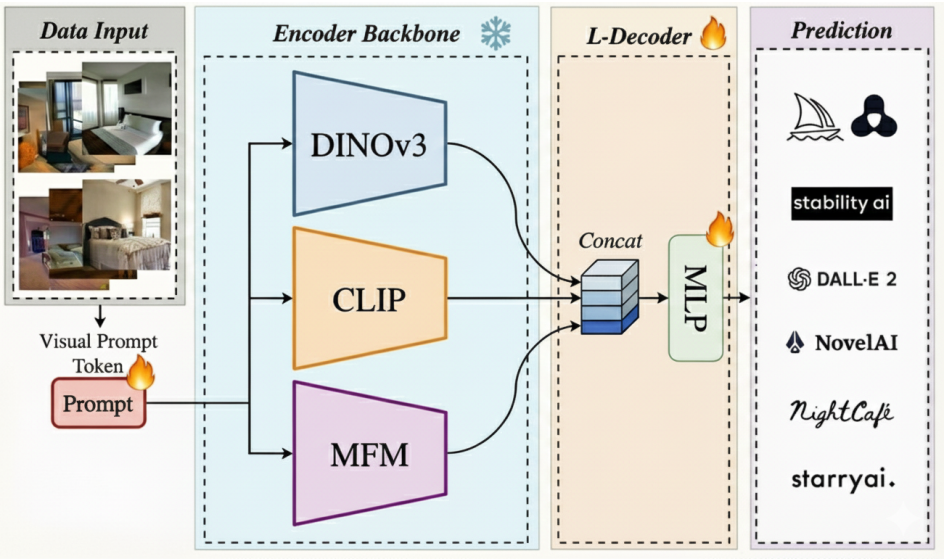

Figure 1. Overall model architecture. input images and visual prompt tokens are injected into frozen pretrained visual encoders selected from DINOv3, CLIP, and MFM to extract complementary visual representations. Depending on the configuration, one or multiple encoders are jointly used, and their output features are concatenated. The aggregated features are then decoded by a lightweight MLP-based L-Decoder into into generator classification predictions

3 Method

This section describes the model configurations and training strategies evaluated for AI-generated image detection and generator identification. We investigate how different visual representations and domain adaptation mechanism contribute to detection performance by systematically combining semantic, visual, and frequency-domain features, as illustrated in Figure 1. Specifically, Section 3.1 presents the model architectures and training strategies used across all evaluated approaches.

3.1 Model Architectures and Training Strategy

Linear Probing Baseline For all linear probing experiments, the pretrained visual encoders are kept frozen and only a lightweight classifier is trained. Specifically, we employ a two-layer multi-layer perceptron (MLP) on top of the extracted image features for generator classification. This setting allows us to evaluate the intrinsic discriminative power of each representation without further modifying the backbone. The DINOv3-Linear Probing baseline utilizes visual features extracted from a pretrained DINOv3 encoder. As a vision-only self-supervised representation, DINOv3 captures visually detailed patterns, structural regularities, and local artifacts without relying on semantic alignment. The extracted features are fed into a two-layer MLP classifier for generator. This configuration is denoted as **DINOv3**. The CLIP-Linear Probing baseline employs the visual encoder of CLIP to extract image representations aligned with natural language supervision. These features encode high-level semantic information. Similar to the DINOv3 baseline, CLIP features are kept frozen and classified using a two-layer MLP. This configuration is denoted as **CLIP**.

CLIP–MFM Feature Fusion To jointly leverage semantic and frequency-domain information, we combine representations from CLIP and Masked Frequency Modeling visual encoder denoted as MFM. Specifically, visual embeddings produced by the CLIP visual encoder and the MFM visual encoder are concatenated at the feature level. The resulting fused representation is then passed to a two-layer MLP classifier. This simple fusion strategy enables the classifier to access complementary high-level semantic cues and low-level frequency-based signals. This configuration is denoted as **CLIP+MFM**.

Visual Prompt Tuning with CLIP. To adapt semantic representations to the generator identification task in a parameter-efficient manner, we employ visual prompt tuning on the CLIP visual

encoder. A deep prompt tuning strategy is adopted, in which learnable prompt tokens are prepended to the input sequence of each transformer layer [8]. During training, only the prompt tokens and the classification head are optimized, while the CLIP backbone remains frozen. This configuration is denoted as **CLIP+VPT**.

CLIP–MFM Feature Fusion with Visual Prompt Tuning Finally, we evaluate a combined setting that incorporates semantic features, frequency-domain cues, and domain adaptation. In this configuration, we apply Visual Prompt Tuning to the CLIP visual encoder and MFM visual encoder to extract prompt-adapted CLIP and MFM features. These features are concatenated and fed into a two-layer MLP classifier. This approach, denoted as **CLIP+MFM+VPT**, aims to exploit complementary information sources while enabling task-specific domain adaptation through prompts.

4 Experiment

In this section, we empirically compare the detection performance of various AI-generated image detection methods explored in our study. Section 4.1 provides an overall evaluation of the compared methods on the generator classification task, highlighting their relative strengths and performance differences. Then, in Section 4.2, we analyze which image generation models are more difficult to detect by examining performance degradation across generator families, with a particular focus on GAN-based and Diffusion-based models.

Dataset. In this study, we use the LSUN Bedroom dataset [9] to systematically analyze AI image generator detection. This dataset is well suited for our task as it is widely used as a benchmark in AI-generated image detection research and provides explicit annotations of image generation models. The dataset includes images generated by 10 different AI image generators, spanning two major categories: GAN-based models and diffusion-based models. The GAN-based generators consist of ProGAN[10], StyleGAN[11], ProjectedGAN[12], Diff-StyleGAN2 and Diff-ProjectedGAN[13], while the diffusion-based generators include DDPM[14], IDDPM[15], ADM[16], PNDM[17], and LDM[18]. This diversity allows us to evaluate detection performance across a broad range of generative mechanisms, from adversarial training to diffusion processes. Each generator provides 10,000 synthesized images, resulting in 100,000 AI-generated images in total.

Due to computational and memory constraints, we restrict our experiments to the validation split of the original dataset. From this validation split, we construct a new dataset by performing an 8:1:1 train–validation–test split, while maintaining class balance across all generator categories. Specifically, each generator contributes an equal number of samples to each split, ensuring a balanced evaluation setting. We focus exclusively on AI-generated images and do not include real images in our experiments, as our objective is to analyze and distinguish generator-specific visual characteristics rather than real-versus-fake classification.

Implementation. All experiments are conducted using a unified training configuration to ensure fair comparison across different combinations. All models are trained for 20 epochs with a batch size of 32. The hidden dimension of the classification head is fixed to 512 across all experiments. For optimization, we use the AdamW optimizer with a learning rate of 1×10^{-4} and a weight decay of 0.01. A cosine annealing learning rate scheduler is applied with $T_{\max} = 10$ to adjust the learning rate during training. For feature extraction, we adopt Vision Transformer (ViT-Base) architectures across all vision encoders to maintain architectural consistency. Specifically, ViT-Base backbones are used for CLIP, DINOv3, and the Masked Frequency Modeling (MFM) vision encoder. For Visual Prompt Tuning (VPT), 20 learnable prompt tokens are prepended to the input token sequence for each transformer layer, while the backbone parameters remain frozen. Unless otherwise specified, all hyperparameters are kept identical across experiments to ensure that performance differences arise from the choice and combination of visual representations.

4.1 Overall Evaluation of Detection Methods

Table 1 summarizes the overall detection performance of the evaluated methods on the generator classification task. Overall, **CLIP** consistently outperform **DINOv3**, indicating that high-level semantic

representations are more effective than fine-grained visual structures for distinguishing between different AI image generators.

When comparing **CLIP** and **CLIP+MFM** configurations, the **CLIP** consistently achieves higher performance. Contrary to our initial expectation that frequency-domain information would provide complementary benefits, incorporating MFM embeddings alongside CLIP embeddings fails to improve performance and instead leads to a noticeable degradation compared to using CLIP alone. In contrast, incorporating visual prompt tuning (VPT) consistently yields substantial performance improvements across different feature configurations. In particular, **CLIP+MFM+VPT** achieves the highest accuracy and F1-score, highlighting the effectiveness of prompt-based adaptation in facilitating generator-specific discrimination. These results indicate that simply concatenating frequency-based representations with semantic embeddings is insufficient for effective generator classification, whereas prompt-based adaptation enables more meaningful integration of complementary cues. Designing more principled fusion strategies between CLIP and MFM embeddings remains an important direction for future work.

4.2 Analysis of Hard-to-Detect Generation Models

To further analyze which types of generation models are more challenging to detect, Table 2 reports the average F1-scores across generator families, comparing GAN-based and Diffusion-based models. Across all evaluated detection methods, performance on Diffusion-generated images is consistently lower than that on GAN-generated images.

Method	Trainable Params	Acc (%)	F1-score (%)
DINOv3	MLP-only	59.3	59.0
CLIP	MLP-only	**83.9**	**84.0**
CLIP+MFM	MLP-only	78.1	78.0
CLIP+VPT	prompt+MLP	**91.5**	**91.0**
CLIP+MFM+VPT	prompt+MLP	**92.1**	**92.0**

Table 1. Accuracy and F1-score performance comparison for 10-way generator classification. First, second, and third places are marked.

Model	GAN avg F1-score (%)	Diffusion avg F1-score (%)
DINOv3	74.8	43.4
CLIP	84.8	82.8
CLIP+MFM	80.2	75.6
CLIP+VPT	95.4	87.4
CLIP+MFM+VPT	94.8	89.0

Table 2. Average F1-score by generator family (GAN vs. Diffusion) for each model setting.

This trend can be attributed to the characteristics of diffusion-based generation, which iteratively refines images through denoising and suppresses localized artifacts commonly exploited by detection models. As diffusion models achieve higher visual fidelity, inter-generator differences in both spatial and frequency domains become increasingly subtle, making discrimination within the diffusion family particularly challenging.

This gap is particularly pronounced for models relying solely on semantic or visually grounded representations. For instance, **DINOv3** experiences a severe performance drop from 74.8% on GANs to 43.4% on diffusion models, and **CLIP** also shows reduced performance (84.8% vs. 82.8%). These results suggest that high-level semantic or visually grounded features alone are insufficient to capture the subtle inter-generator differences within the diffusion family. Notably, incorporating visual prompt tuning (VPT) leads to a significant improvement, with **CLIP+VPT** achieving a diffusion F1-score of 87.4%,

demonstrating that lightweight adaptation enables pretrained representations to better emphasize generator-specific cues.

Importantly, Table 2 further reveals that the strongest discrimination among diffusion models is achieved when visual prompt tuning is combined with frequency-domain representations. While adding frequency features alone yields moderate gains (**CLIP+MFM**: 75.6%), their integration with VPT results in the highest diffusion F1-score of 89.0% (**CLIP+MFM+VPT**), outperforming all other configurations. This indicates that frequency-domain cues provide complementary information that becomes most effective when the model is explicitly guided to attend to task-relevant patterns via prompt tuning. Together, these findings demonstrate that jointly leveraging adaptive visual prompting and frequency-encoded representations is critical for robustly distinguishing between diffusion-based generators.

5 Limitations and Future Work

Despite promising results, our study has several limitations that suggest important directions for future work. First, our current approach adopts a relatively simple fusion of MFM-based frequency features and CLIP visual embeddings, which may not fully capture their complementary interactions; more advanced fusion strategies, such as adaptive weighting or attention-based mechanisms, could further improve robustness and discriminative power. In addition, our experiments are conducted on a fixed set of AI generators, limiting the model's ability to generalize to real images or unseen generation models. Extending the framework to zero-shot or open-set scenarios, where the detector can identify real images and previously unseen generators without explicit supervision, is an important next step. Finally, our evaluation is based on a dataset with limited scale and category diversity; expanding to larger datasets covering a wider range of objects, scenes, and real-world contexts would enable more comprehensive evaluation and improve scalability. Addressing these limitations would move our work toward a more practical and generalizable AI-generated image detection framework for real-world deployment.

6 Conclusion

In this paper, to address the rapid evolution of AI image synthesis, we pioneer the task of **AI Image Generator identification** beyond conventional binary real/fake detection—identifying the specific generative model responsible for a given synthetic image. By analyzing the unique visual fingerprints left by generative models, we explore the potential for enhanced accuracy in digital forensics and the establishment of safety in the era of generative AI.

Our extensive experiments demonstrate that classification performance is maximized when combining CLIP, which captures high-level semantic styles, with MFM (Masked Frequency Modeling), which traces imperceptible digital artifacts adopted with domain adaptation method(VPT). In particular, the CLIP+MFM+VPT configuration achieved the highest performance, yielding an accuracy of 92.1. This approach, integrating semantic and frequency-domain information, proved to have robust discriminative power across diverse generative architectures, including both GANs and Diffusion models. The inclusion of Visual Prompt Tuning (VPT) was instrumental in this success; it allowed the model to effectively capture subtle, generator-specific cues that are often suppressed in Diffusion models, providing a parameter-efficient way to adapt pretrained features for precise without the need for full fine-tuning.

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